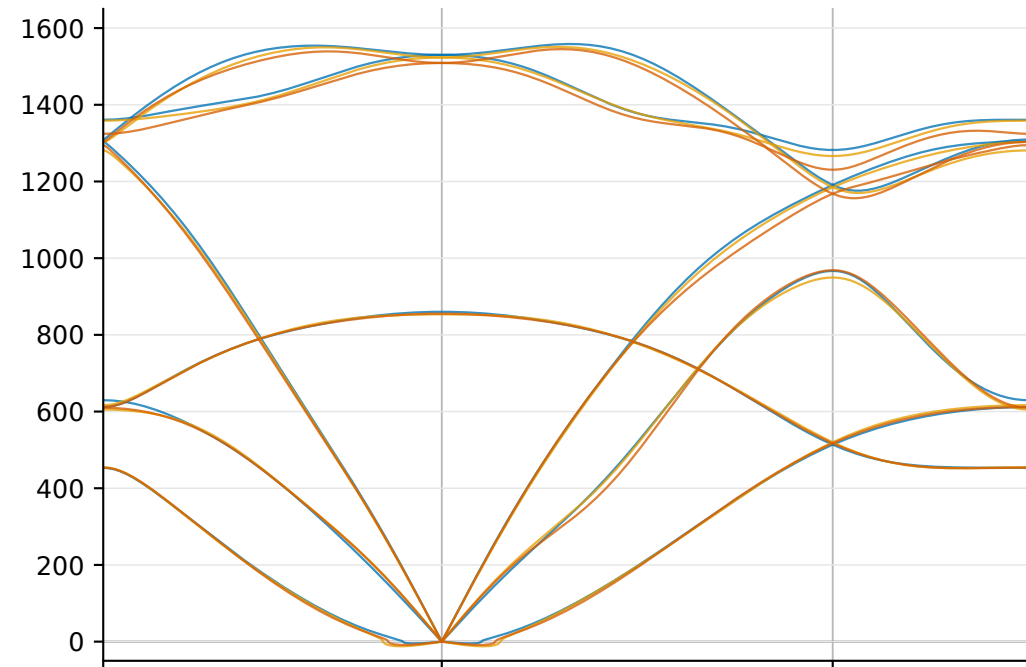
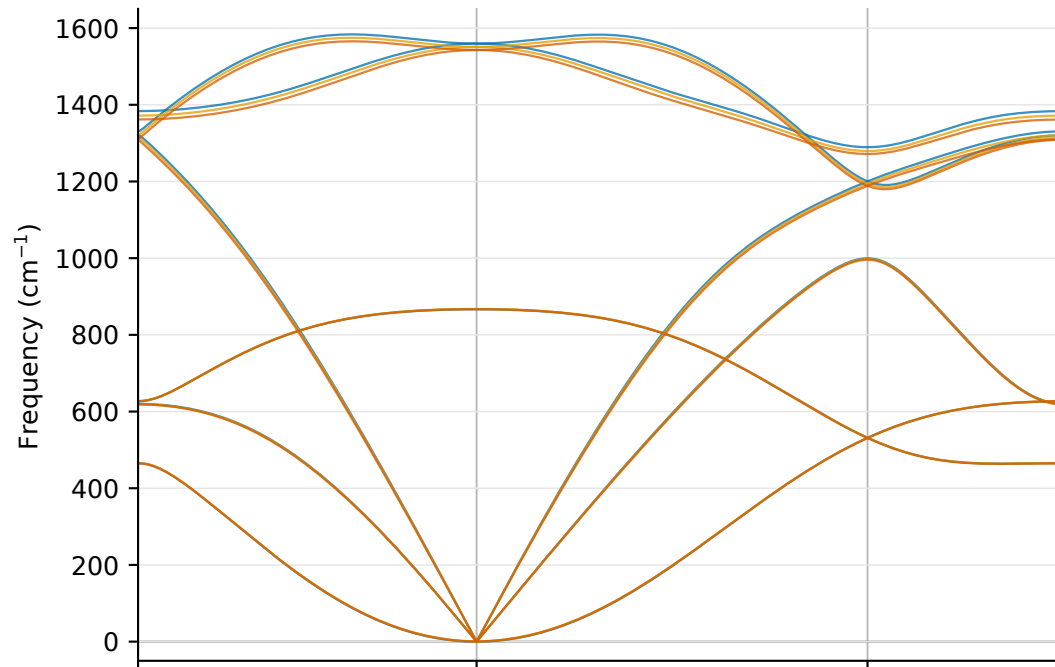


Graphene phonon spectra under diagonal physical-temperature conditions

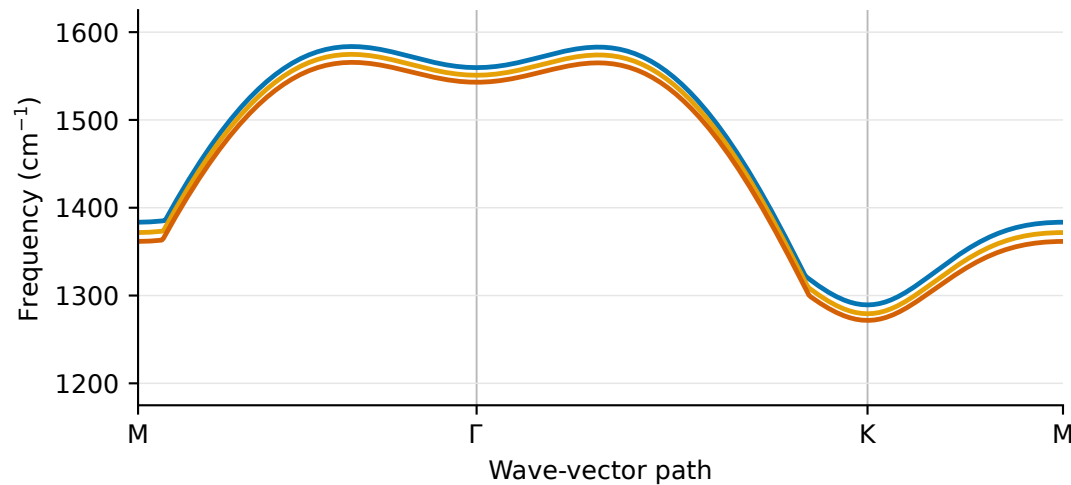
The lattice temperature and the matched electronic smearing operator vary together.

(a) Classical MD/TDEP (L0)

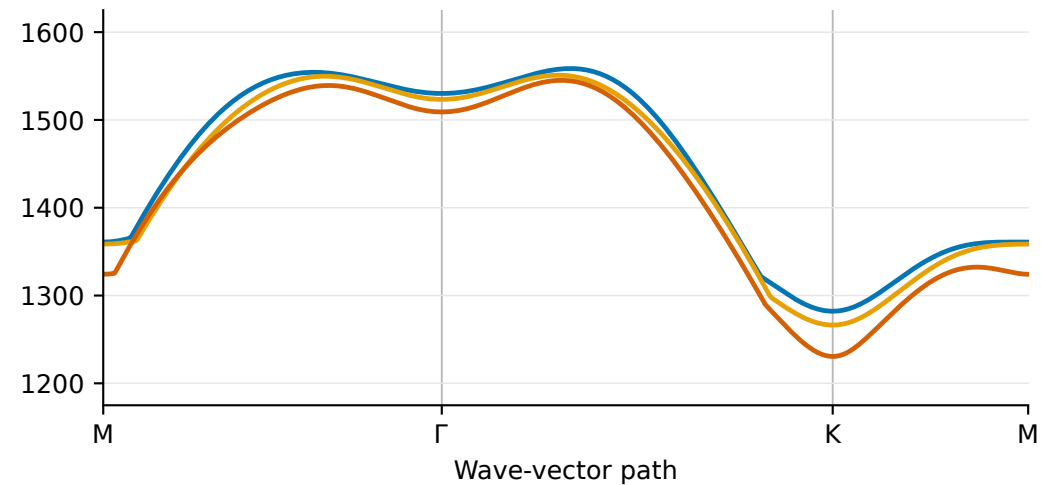
(b) Quantum SSCHA (Q0)



Highest optical branch



Highest optical branch



lattice temperature / smearing/degauss
— 300 K / 0.00190 Ry — 450 K / 0.00285 Ry — 600 K / 0.00380 Ry